Masters in Management
T2 – January to March, 2023

Strategy In The Age Of AI

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COURSE DESCRIPTION

Machine Learning (ML), and Artificial Intelligence (AI) more broadly, is reshaping the way organizations operate and compete. These new technologies and tools reinforce the existing and create new sources of competitive advantage for firms. In the process, the way that companies compete and create value changes as well which has profound implications for business strategy. In this course we will look at how AI technologies are transforming industries, companies, and jobs and how these tools can help companies create and capture value. We will examine how AI technologies change the nature of competition and unlock new possibilities for strategic decision makers.

This course recognizes that understanding the interface between strategy and AI technologies is especially important given that an increasing number of companies declare that AI technologies are critical to their business. In a survey conducted by ESSEC Business School in 2019 among companies operating in France, over two thirds of respondents considered AI to be an important or very important factor for their business strategy. Similarly, according to a report by Deloitte published in 2020, 73 percent of survey respondents believed that AI was very or critically important to their business. Adoption of AI is also progressing. According to a recent McKinsey survey on the State of AI, as of 2020, over half of respondents reported adopting AI in at least one function. Over the past few years ML and AI has moved from the world of science fiction and into the boardrooms of companies across the world making it a key area for business school graduates to master.

Throughout this course you will learn several key concepts of business strategy and how they connect to and benefit from AI. This course offers a mix of lectures, group exercises, and case studies, where we will examine and discuss how companies have been dealing with the opportunities and threats presented by automation, AI, and ML technologies. You will also have an opportunity to work with data and use machine learning tools to uncover new sources of competitive advantage and value.

While no prior knowledge of programming or data science is required, several simple exercises will help you to build a better intuition for what AI and ML can and cannot provide. The goal is not to turn you into a data scientist—this is a strategy course after all—but to be able to intelligently engage with data scientists in

pursuit of new applications of AI and ML in a business. The growing realization in the business community is that most AI and ML projects fail not because of technology but because of a lack of understanding of how to intelligently apply these new technologies in areas where they can create the most value. In other words, it is the managers' lack of sufficient understanding of AI and ML and how they fit into their business that is hampering the efforts to leverage these technologies. This course aims at improving this understanding.

IMPORTANT: Prerequisites

- Students are required to complete the Strategy and Management core course (MGTS 31103).
- Knowledge of programming language like Python is recommended, but not required.

GENERAL COURSE OBJECTIVES

The general goal of this course is to create a better understanding of how AI technologies can help businesses to achieve and maintain sustainable competitive advantage and thus create value and to train future managers who will lead in AI-powered organizations. We will discuss such questions as:

1. What is AI and ML in particular? What are they for? What are their strengths and limitations from the strategy perspective?
2. What is the impact of AI and ML on industries, companies, and competition?
3. How can AI and ML help companies to create more value and improve their strategies?
4. What is the role of managers in the age of automation and AI?

The course will follow three general areas of discussion—focusing on a big picture first, and later diving into specific subjects related to companies and individuals. The three general areas are: Industry & Technology, Company, and Individual. The topics discussed in each area are presented in the table below.

INDUSTRY & TECHNOLOGY	COMPANY	INDIVIDUAL
• Explore the ways in which AI and ML are transforming industries and competition • Understanding and navigating the AI and ML competitive disruption • What are the key AI and ML technologies from the strategy perspective	• Which areas in business will benefit most from AI and ML • How to make AI and ML the core capability of a firm • How to successfully choose and launch the first AI pilot in a company and overcome organizational resistance • How to organize a company for AI future - the role of organization design and culture	• What are the key managerial skills in the age of AI • How to build and lead effective teams in AI-powered organizations • Managing human capital in the age of AI • How to stay relevant and have a great career in the 21st century

TEXTBOOKS

While there is no specific textbook for the course, we recommend to use the following books to give further context for the subjects discussed in class:

- Iansiti M, Lakhani KR. 2020. Competing in the age of AI: strategy and leadership when algorithms and networks run the world. Harvard Business Press

- Davenport T, Harris J. 2017. Competing on analytics: Updated, with a new introduction: The new science of winning. Harvard Business Press

Additional books, websites, and articles

- Agrawal A, Gans J, Goldfarb A. 2020. How to win with machine learning. Harvard Business Review (September-October): 126–133.
- Hagiu A., Wright J. 2020 When data creates Competitive Advantage, Harvard Business Review
- Maini V. 2018, May 28. A Beginner's Guide to AI/ML. Medium. Available at: https://medium.com/machine-learning-for-humans/why-machine-learning-matters-6164faf1df12 [10 February 2021].
- Ng A. 2019. Chapter 6: Pick a quick win to build internal support. In Artificial Intelligence: The Insights You Need from Harvard Business Review. Harvard Business Press.
- Ransbotham S, Kiron D, Gerbert P, Reeves M. 2017. Reshaping business with artificial intelligence: Closing the gap between ambition and action. MIT Sloan Management Review. Massachusetts Institute of Technology, Cambridge, MA 59(1).
- Tambe P, Cappelli P, Yakubovich V. 2019. Artificial intelligence in human resources management: Challenges and a path forward. California Management Review. SAGE Publications Sage CA: Los Angeles, CA 61(4): 15–42.
- Wilson HJ, Daugherty PR. 2018. Collaborative intelligence: humans and AI are joining forces. Harvard Business Review 96(4): 114–123.

A FEW SIMPLE RULES

Much of the learning comes from class discussions. You should come to class prepared to engage fully with your peers and professor. This means that you are expected to:

- Arrive on time. Late arrivals will miss the quiz and thus associated points.
- Remain in class throughout the session
- Be prepared and ready to question and debate, while always behaving in a professional and respectful manner toward your peers and professor.
- You are expected to work out issues of co-operation and division of work within the team, and to arrange for equitable contribution of teamwork among yourselves. If severe issues emerge during the term in your group, please contact me as soon as possible.

COURSE OUTLINE

The course will consist of 10 sessions. The actual layout of the sessions, cases, exercises, and assigned readings are subject to change and will be updated.

I. Industry & Technology

Session 1	**What Is AI And How Does It Affect Competitive Advantage?**
Case:	Intel: AI and Industry 4.0 Strategy - please read the case before class and be ready to answer the following questions: • How did Industry 4.0 impact operating environments in companies? • How did Intel's AI solutions differ from other available options? • How can Intel sell the value of its AI approach to clients?
Readings:	"Who Profits From Industry 4.0?", Helper, Martins, Seamans (2019), NBER
Video:	https://www.youtube.com/watch?v=HPRURtORnis

Session 2	**How AI Is Transforming Industries**
Case:	Preferred Networks - please read the case before class and be ready to answer the following questions: • What are the key technologies about? What problems do they solve? • What possible business models (ways of making money) can the company consider? • What are the strengths and limitations of PFN's technologies?
In-class exercise:	Short group project and presentations. Details will be revealed during the class.
Video:	Machine, Platform, Crowd, interview with A. McAfee and Erik Brynjolfsson https://www.youtube.com/watch?v=ofeAFgU_hGQ

Session 3	**Non-Market Strategies And AI Adoption**
Article:	General Motors and Autonomous Vehicle Regulations - please read the article before class and be ready to answer the following questions: • Is the pending federal regulation likely to be good or bad news for General Motors? • Should General Motors attempt to shape the legislation or influence the timing of its implementation? • What would be the desired regulatory outcomes?
In-class exercise:	Machine Learning Workshop
Readings:	"Hall of Mirrorrs", Bertrand et al. (2018), NBER

Session 4	**Industry And AI: Group Presentations And A Guest Speaker**
In-class exercise:	Group presentations
Fireside chat:	Guest talk. AI and Industry Evolution

II. AI, Strategy, and Organization

Session 5	**Creating Blue Oceans With The Help Of AI**
Case:	Stitch Fix - please read before class and be ready to discuss the following questions: • Katrina Lake looked across three industries to create her new market space: retail, personal stylist, and subscription boxes. Which elements did she take from these industries and which did she leave behind? • How can Lake keep growing her Blue Ocean and protect it from other players, like Amazon?
In-class exercise:	Machine Learning Workshop
Readings:	To be announced

Session 6	**AI Transformation Of An Organization**
Case:	The future of patent examination at the USPTO - please read the case before class and consider the following questions: • At what speed should the machine learning tool Sigma be introduced? • Who are the key stakeholders that Hirshfeld should approach and what message should he communicate? • What metrics should he consider to assess patent examiners' performance?
In-class exercise:	A guest speaker – topic to be announced later.
Readings:	To be announced

Session 7	**Responsible AI: Dealing With Bias And Privacy Issues**
Case:	Programming a "Fairer" System - please read the case before class and be ready to answer the following questions: • How would you go about evaluating the fairness of COMPAS software? • How would you protect the intellectual property related to your AI software? • What are the potential risks and benefits of the AI system?
In-class exercise:	Group work *(to be discussed in class)*
Readings:	"Sporting Chances", Leta-Jones, Levy (2018), SSRN

III. Teams and Individuals

Session 8	**AI And The Future Of Work**
Case:	Unilever's Response to the Future of Work - please read the case before class and be ready to answer the following questions: • How is the commerce sector changing and what does that require from Unilever? • How do you evaluate Unilever's Future of Work program? • How well does Unilever balance purpose and automation with its approach?

In-class exercise:	"Your Purpose Workshop"
Readings:	To be announced

Session 9	**Human Intelligence: The New Superpower Of The AI Age**
Case:	• None this time
In-class exercise:	"Discovering your own deep goals to improve decision making."
Readings:	"Algorithms need managers too" Luca et al. (2016) HBR "Leaders as decision architects" Beshears & Gino (2015) HBR

IV. Final Project Presentations

Session 10	**Final Group Presentations**
In-class exercise:	Using AI to Build a Better Business School of Tomorrow

EVALUATION

The grading scheme is as follows:

- Group assignments and exercises **(20 % of total grade)**
- Class participation in discussions **(30% of total grade)**
- Individual quizzes **(15% of total grade)**
- 360 degree evaluation by other team members **(10% of total grade)**
- Final report by group and final presentation **(25 % of total grade)**

Late submissions of any assignment deliverable will not be accepted.

Grading scale

18 – 20	Excellent	11 – 12	Satisfactory
15 – 17	Very Good	10	Pass
13 – 14	Good	< 10	Fail

You are expected to attend all classes. By missing one session, you create gaps in your knowledge that compromise your ability to follow (and enjoy) subsequent sessions. If for some unavoidable reason, you must miss a class, it is your responsibility to find out from your classmates what material was covered and what assignments or handouts you may have missed. ***A maximum of two absences are allowed.***

In this class you will be graded on a z-curve, where your individual performance will be compared relative to the rest of the class. As a result, it is not easy to receive maximum points in a class where you are surrounded by very smart students. While with such a scheme it is impossible for everyone to get above-average grades,

you will receive a better estimate of your personal competitive advantage and your strengths and weaknesses – an important step in improving your personal value.

ESSEC PLAGIARISM POLICY

All deliverables should be your own work. When building on the ideas of others, it is important to cite authors' work in a reference section or in a footnote. Using text that has appeared elsewhere (both print and electronic form) is indicated by quotation marks and a citation to that source. School policy states that failing to do the above constitutes plagiarized work that will result in the student or group receiving a "0" for the course. Please see the Student Handbook "Commitments and Responsibilities" section for details.

INSTRUCTOR

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I am an Associate Professor of Management at ESSEC, where I teach strategy and strategic management courses in the school's Masters in Management, GMBA, and Executive Programs as well as strategy, organization theory, computational modeling, and machine learning in the school's doctoral program. In my research I focus on how organizational structure influences the way companies adapt to industry changes and innovate. My research has been published in the Strategic Management Journal, Organization Science, Journal of Organization Design and the Journal of Management Inquiry. Before joining ESSEC in 2016, I did my MBA at INSEAD, and later earned a PhD in Management (Strategy) from the same school. Prior to INSEAD, I worked at Deloitte in Canada and at Siemens AG in Germany.